
Depression

Learning Objectives

- **Describe** neurochemical and neurotrophic theories of affect and their modulation by antidepressants.
- **Identify** major antidepressant classes and their cellular targets.
- **Apply** antidepressant therapy to common non-depressive disorders and conditions.
- **Compare** adverse effects and their mechanisms across antidepressant classes.
- **Recognize and manage** serotonin syndrome.



ADRENALINE/EPINEPHRINE

fight or flight
Produced in stressful situations. Increases heart rate and blood flow, leading to physical boost and hightened awareness.

GABA

calming
Calms firing nerves in the central nervous system. High levels improve focus, low levels cause anxiety. Also contributes to motor control and vision.

NORADRENALINE/NOREPINEPHRINE

concentration
Affects attention and responding actions in the brain. Contract blood vessels, increasing blood flow.

ACETYLCHOLINE

learning
Involved in thought, learning, and memory. Activates muscle action in the body. Also associated with attention and awakening.

DOPAMINE

pleasure
Feelings of pleasure, also addiction, movement and motivation. People repeat behaviors that lead to dopamine release.

GLUTAMATE

memory
Most common neurotransmitter. Involved in learning and memory, regulates development and creation of nerve contacts.

SEROTONIN

mood
Contibutes to well-being and happiness. Helps sleep cycle and digestive system regulation. Affected by exercise.

ENDORPHINS

euporia
Released during exercise, excitement and sex. Producing a sense of well-being and pain reduction.

Disorders associated with neurotransmitters

Alzheimer’s disease

Alzheimer’s disease is a neurodegenerative disorder characterized by learning and memory impairments. It is associated with a lack of **acetylcholine** in certain regions of the brain.

Depression

Depression is believed to be caused by a depletion of **norepinephrine, serotonin, and dopamine** in the central nervous system.

Schizophrenia

Schizophrenia, which is a severe mental illness, has been shown to involve excessive amounts of **dopamine** in the frontal lobes, which leads to psychotic episodes in these patients.

Parkinson’s disease

The destruction of the substantia nigra leads to the destruction of the only central nervous system source of **dopamine.**

Epilepsy

Some epileptic conditions are caused by the lack of inhibitory neurotransmitters, such as **GABA,** or by the increase of excitatory neurotransmitters, such is **glutamate.**

Huntington’s disease

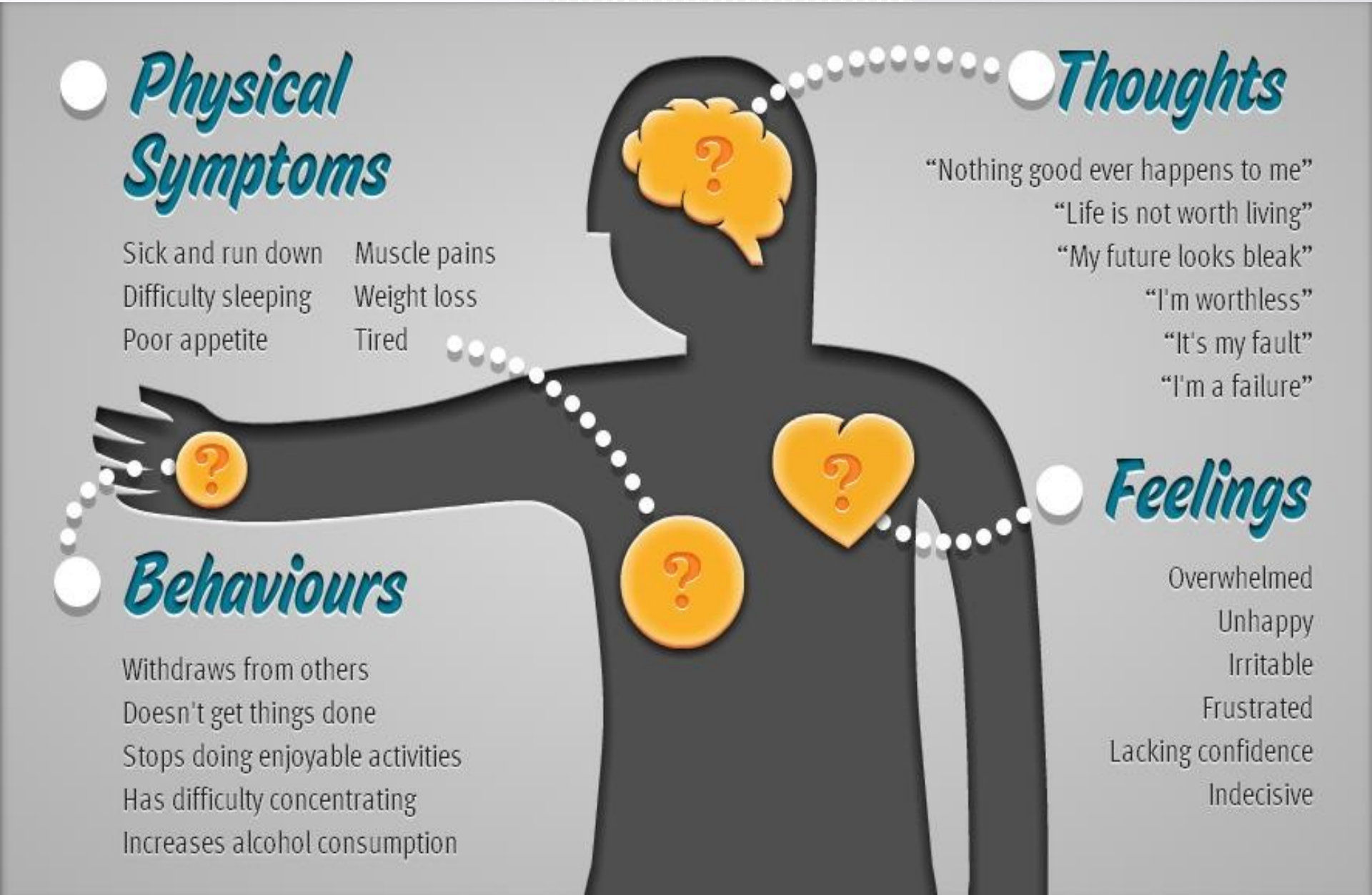
Besides epilepsy, a chronic reduction of **GABA** in the brain can lead to Huntington’s disease.

Depression

- common clinical condition associated with depressed mood most of the time for **at least 2 weeks** and/or loss of interest or pleasure in most activities

- In addition, depression is characterized by disturbances in sleep and appetite as well as deficits in cognition and energy. Thoughts of guilt, worthlessness, and suicide are common.

Clinical presentation



Pathophysiology of Depression

Monoamine Hypothesis

- deficiency in amount/function of cortical and limbic:
  - Serotonin(5HT)
  - Norepinephrine(NE)
  - Dopamine(DA)

Neurotropic Hypothesis

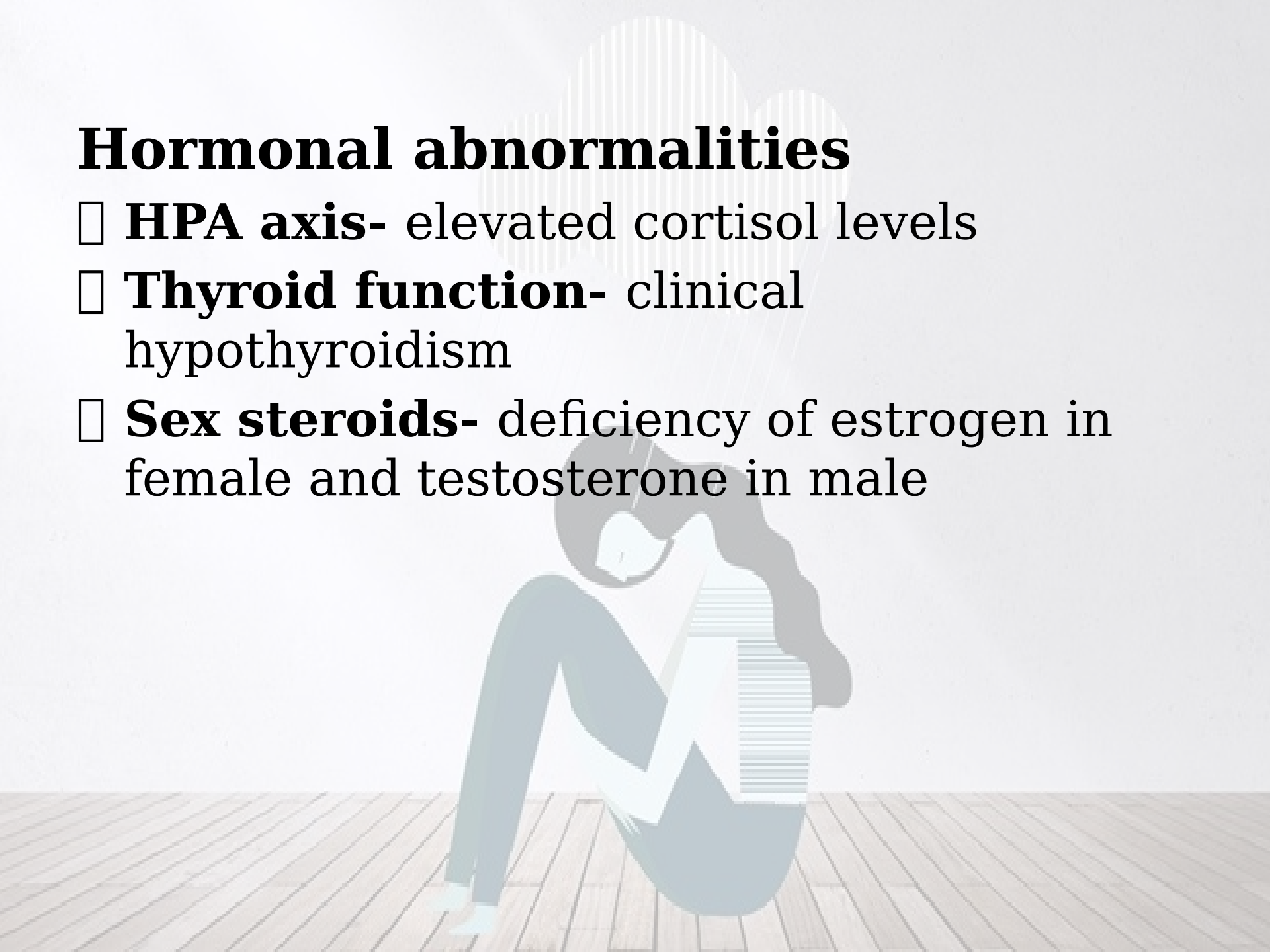
- Decrease in the brain derived neurotropic factors(BDNF)

Glutamate Hypothesis

- Elevated levels of glutamate in CSF

Hormonal abnormalities

- **HPA axis-** elevated cortisol levels
- **Thyroid function-** clinical hypothyroidism
- **Sex steroids-** deficiency of estrogen in female and testosterone in male



Types of Depression

- Major Depressive Disorder
- Persistent Depressive Disorder
- Bipolar Disorder
- Seasonal Affective Disorder (SAD)
- Psychotic Depression
- Postpartum Depression
- Premenstrual Dysphoric Disorder (PMDD)
- Situational Depression
- Atypical Depression

Major Depression



- Mostly occurs suddenly, possibly triggered by loss, crisis or chang
- Clinically diagnosed as major depression, if symptoms persist for more than 2 weeks
- Continue for months or years

Persistent Depressive Disorder

- Chronic, low grade depressed mood without sufficient symptoms to label as major depression
- Similar symptoms but less severe
- Symptoms usually begin two years before diagnosis



Bipolar Disorder



- Mood changes episodically between periods of depression and mania
- High’s and Low’s
- Low phase, symptoms of major depression seen



Seasonal Affective Disorder (SAD)

- SAD : mood disorder that has seasonal pattern

- Cause of disorder unclear; however thought to be related to variation in light exposure in different seasons

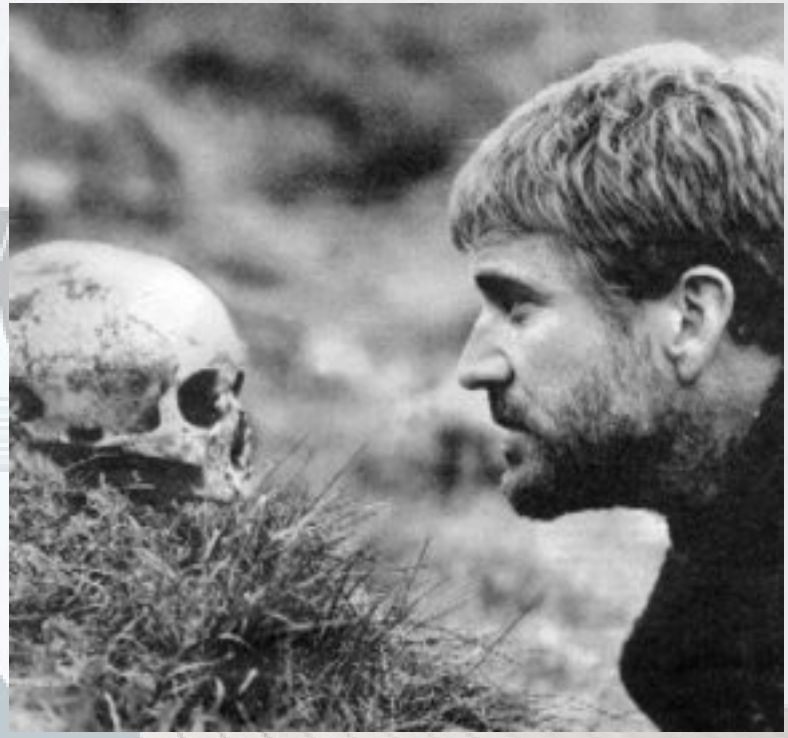
- Most common in winter





Psychotic Depression

- have the symptoms of major depression along with "psychotic" symptoms as:
- Hallucinations
- Delusions
- Paranoia



Postpartum Depression

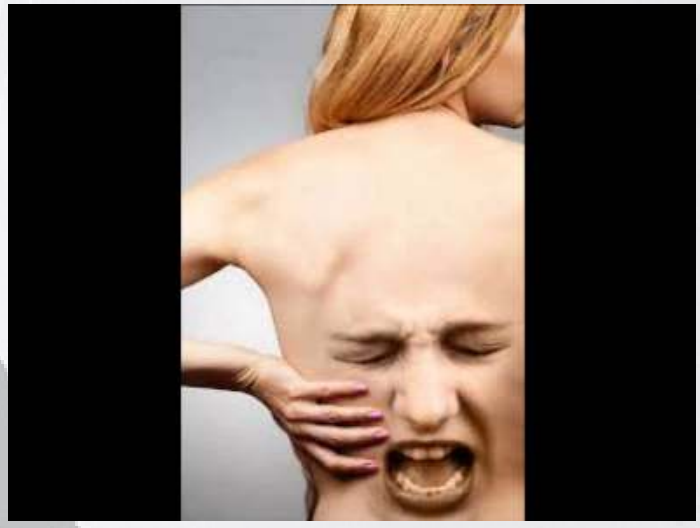
- Women- weeks and months after childbirth may have postpartum depression.



Premenstrual Dysphoric Disorder (PMDD)

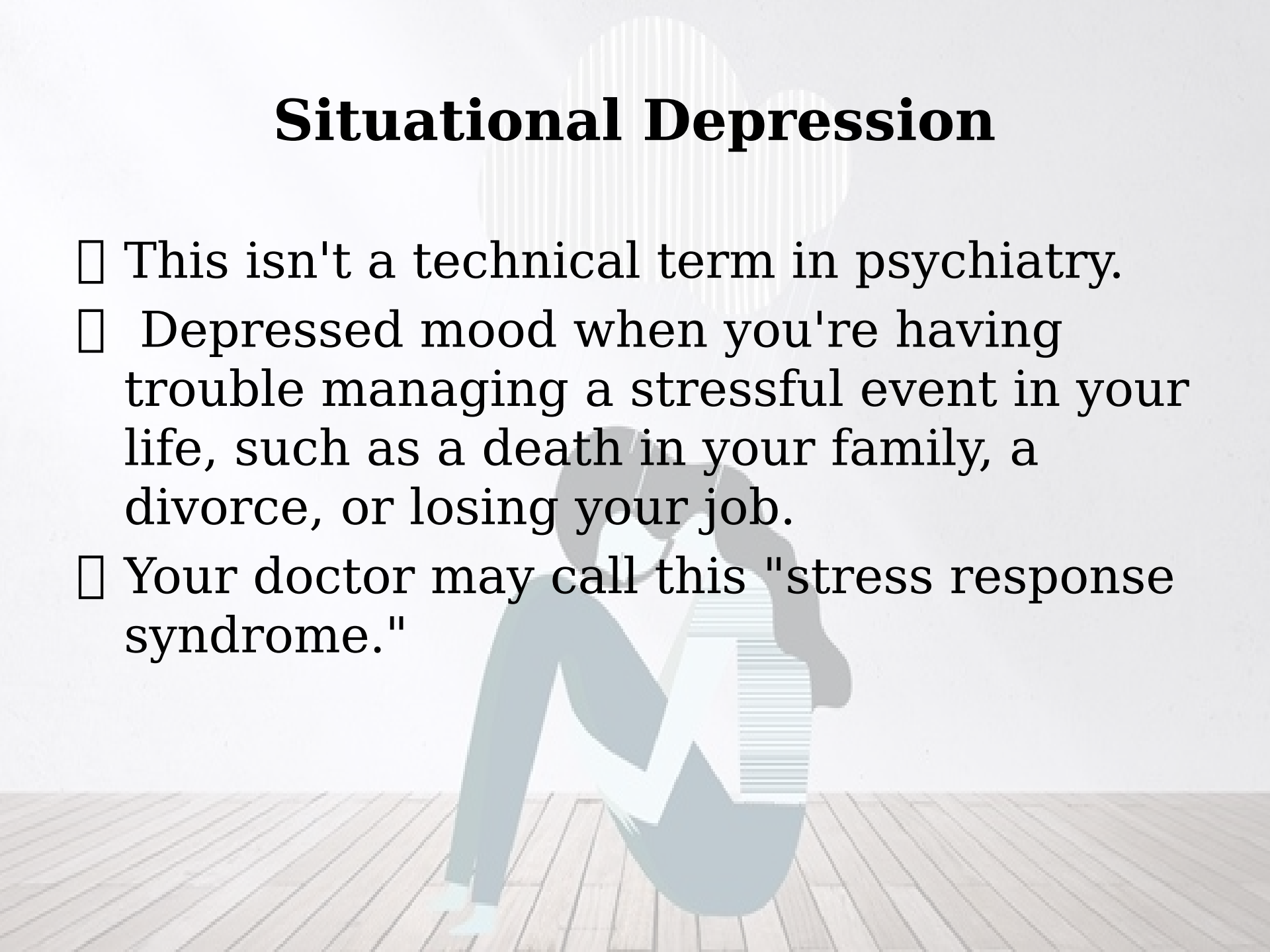
Depression and other symptoms at the start of their period:

- Mood swings
- Irritability
- Anxiety
- Trouble concentrating
- Fatigue
- Change in appetite or sleep habits



Situational Depression

- This isn't a technical term in psychiatry.
- Depressed mood when you're having trouble managing a stressful event in your life, such as a death in your family, a divorce, or losing your job.
- Your doctor may call this "stress response syndrome."

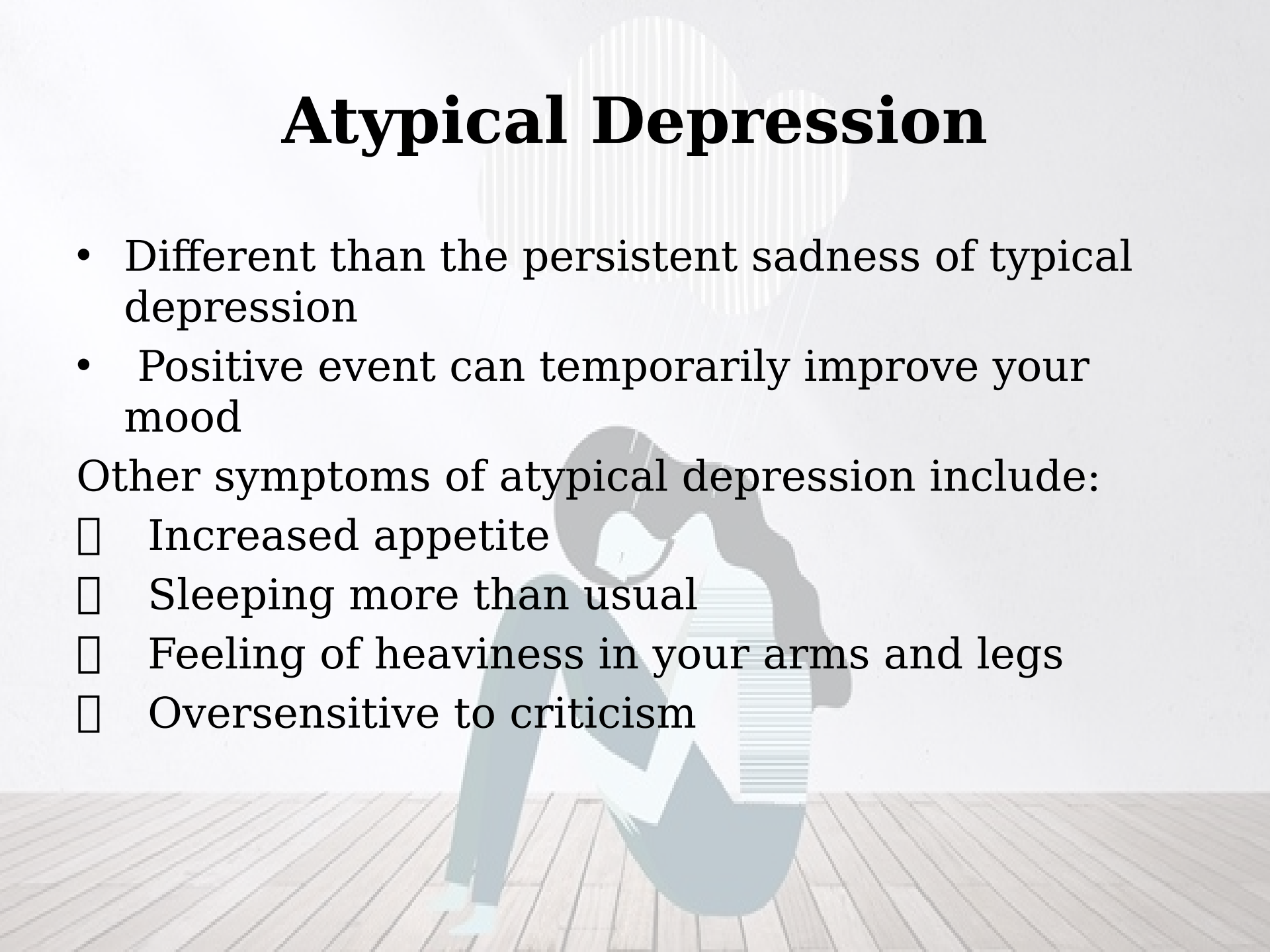


Atypical Depression

- Different than the persistent sadness of typical depression
- Positive event can temporarily improve your mood

Other symptoms of atypical depression include:

- Increased appetite
- Sleeping more than usual
- Feeling of heaviness in your arms and legs
- Oversensitive to criticism



How to Diagnose?

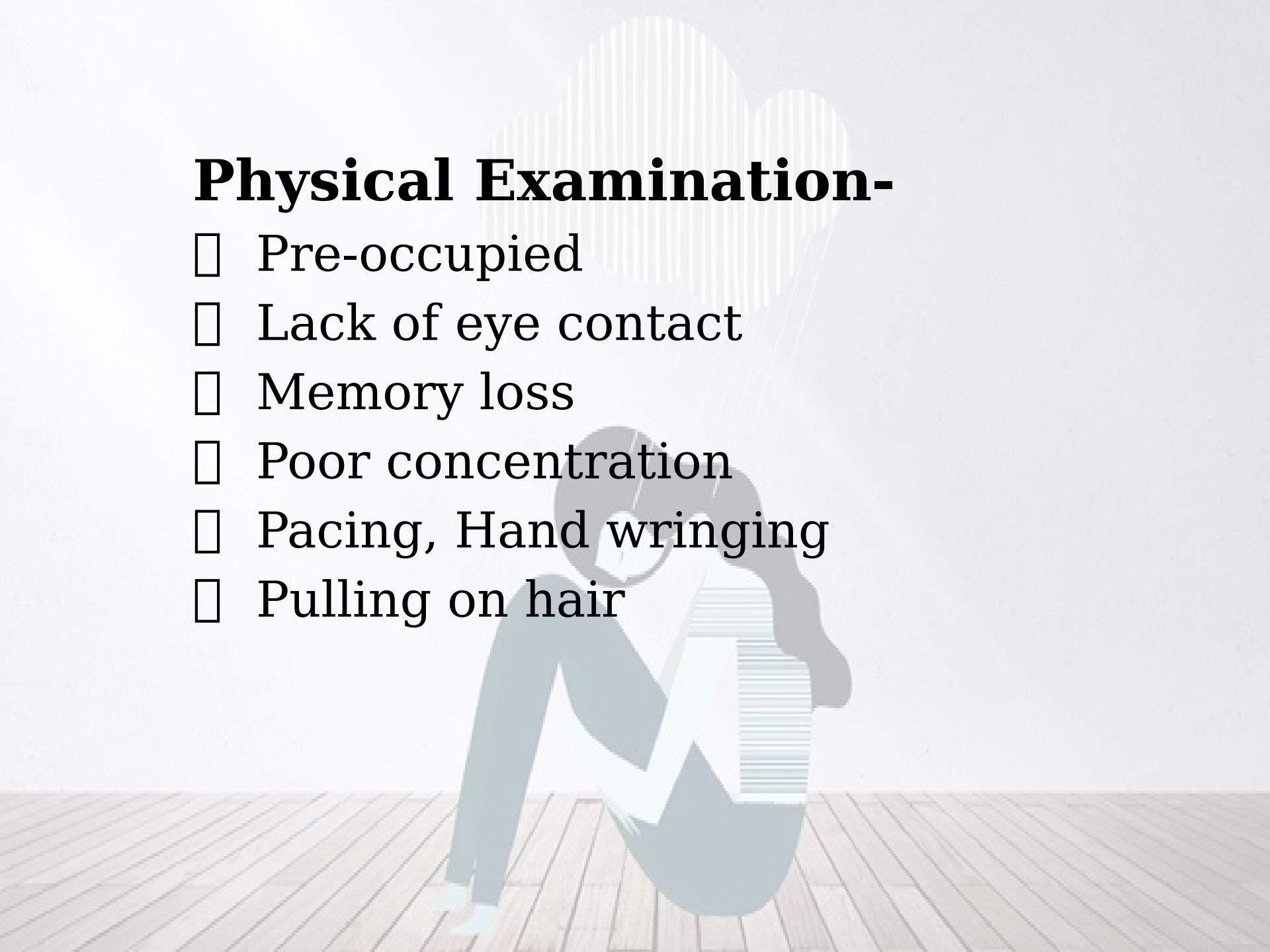
- Laboratory tests- Not helpful
- History Taking





Physical Examination-

- Pre-occupied
- Lack of eye contact
- Memory loss
- Poor concentration
- Pacing, Hand wringing
- Pulling on hair



TESTS

- Beck Depression Inventory (BDI)
- Zung Self-Rating Depression Scale
- Hamilton Rating Scale for Depression (HRSD)



Management

Psychotherapy

Cognitive Behavioral Therapy (CBT) is one of the most common types of talk therapy used in the treatment for depression. CBT focuses on

- **Education** (pinpointing the life problems that contribute to a patient's depression & which aspects of those problems they may be able to solve or improve)
- **Improving self-esteem** (a patient's sense of self-regard, confidence & hope)
- **Psychological (ego) functioning** (cognitive abilities, affect regulation, morals & ability to relate to others)
- **Adaptive skills** (behaviors associated with effective function with friends, family & coworkers

Antidepressant Drugs

SELECTIVE SEROTONIN REUPTAKE INHIBITORS (SSRIs)

Citalopram CELEXA
Escitalopram LEXAPRO
Fluoxetine PROZAC
Fluvoxamine LUVOX CR
Paroxetine PAXIL
Sertraline ZOLOFT

SEROTONIN/NOREPINEPHRINE REUPTAKE INHIBITORS (SNRIs)

Desvenlafaxine PRISTIQ
Duloxetine CYMBALTA
Levomilnacipran FETZIMA
Venlafaxine EFFEXOR

ATYPICAL ANTIDEPRESSANTS

Bupropion WELLBUTRIN, ZYBAN
Mirtazapine REMERON
Nefazodone
Trazodone DESYREL
Vilazodone VIIBRYD
Vortioxetine BRINTELLIX

TRICYCLIC ANTIDEPRESSANTS (TCAs)

Amitriptyline
Amoxapine
Clomipramine ANAFRANIL
Desipramine NORPRAMIN
Doxepin SINEQUAN
Imipramine TOFRANIL
Maprotiline LUDIOMIL
Nortriptyline PAMELOR
Protriptyline VIVACTIL
Trimipramine SURMONTIL

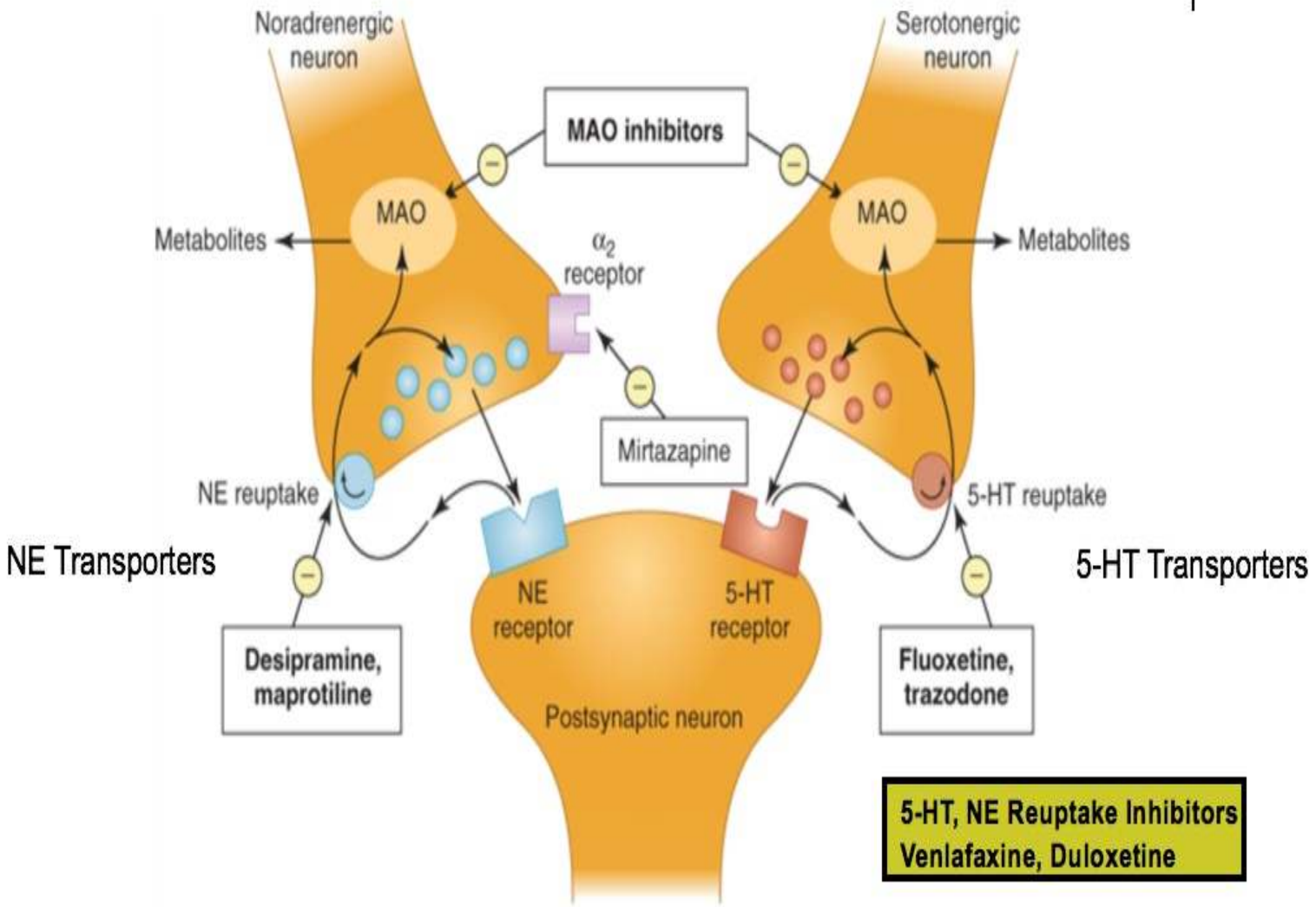
MONOAMINE OXIDASE INHIBITORS (MAOIs)

Isocarboxazid MARPLAN
Phenelzine NARDIL
Selegiline EMSAM
Tranylcypromine PARNATE

Classification of Antidepressants

1. **Tricyclic anti-depressants (TCAs)**
   Imipramine, desipramine, nortriptyline, protryptyline, amytriptiline, doxepin.
2. **Monoamine oxidase inhibitors (MAOIs).**
   Moclobemide, tranylcypromine.
3. **Selective serotonin reuptake inhibitors (SSRIs)**
   Fluoxetine, sertraline, paroxetine,
4. **Serotonin/Norepinephrine reuptake inhibitors**
   Duloxetine, Venlafaxine
5. **Atypical anti-depressants**
   Bupropion, Mianserin, Mirtazepine, Trazodone

Antidepressants: MOA



Selective serotonin reuptake inhibitors (SSRIs)

Fluoxetine, **S**ertraline, **P**aroxetine, **C**italopram,**E**scitalopram

MOA-

- Inhibits the serotonin transporter(SERT) → block reuptake of 5HT into the neuron → increases the availability of 5HT receptors in the CNS enhances serotonergic activity

Pharmacokinetics

- well absorbed from gut
- primarily metabolized by hepatic CYP450 enzyme
- The elimination half-life for most SSRIs is 20-30 hrs
- Fluoxetine metabolism yields an active metabolite (**norfluoxetine**) (long half-life of 4-16 days)

Indications- major depressive disorders, anxiety, obsessive compulsive disorder, posttraumatic stress, premenstrual dysphoric disorder, bulimia nervosa, binge eating disorder

Adverse effects-

- GI disorders- nausea, diarrhea, heartburn
- Sexual dysfunction- loss of libido, diminished arousal
- Headache, anxiety, insomnia
- Weight gain- paroxetine
- Discontinuation syndrome
- Citalopram-can prolong QT interval in a dose-dependent manner

Drug interactions-

SSRIs can produce **drug interactions** due to inhibition of P450-enzymes that metabolize other drugs

Serotonin syndrome-when used in combination with other serotonergic agents produces tremors, rigidity, sweating, hyperthermia, seizures and coma due to increase levels of

Amphetamines	Analgesics	Antidepressants	Antiemetics	Antimigraine	Miscellaneous
Dextroamphetamine	Cyclobenzaprine	MAO inhibitors	Metoclopramide	Carbamazepine	Buspirone
Methamphetamine	Fentanyl	SSRIs	Ondansetron	Ergot alkaloids	Cocaine
	Meperidine	SNRIs		Triptans	Dextromethorphan
	Tramadol	Trazodone			Levodopa
		St John's wort			Linezolid
		TCAs			Lithium
					MDMA (Ecstasy)

Serotonin Norepinephrine reuptake inhibitors (SNRIs)

Venlafaxine, **D**esvenlafaxine, **D**uloxetine, **M**ilnacipran

MOA-

Inhibits the reuptake of serotonin and noradrenaline into the neuron

No anticholinergic, antihistaminic,

Adverse effects-

Same to SSRIs

Additional noradrenergic related effects- increased BP and heart rate

Others- insomnia and anxiety

Duloxetine- hepatotoxicity

Venlafaxine- cardiac toxicity

Discontinuation syndrome

Tricyclic antidepressants (TCAs)

Amitriptyline, Nortriptyline, Desipramine, Imipramine

MOA-

- Inhibits SERT → block reuptake of 5HT into neuron and enhance the availability of 5HT at the receptors in the CNS
- Inhibits NET → block reuptake of NA into neuron and enhance the availability of NA at the receptors in the CNS
- Also blocks alpha adrenergic receptors(α1), histamine(H1) and muscarinic(M1)

Adverse effects:

- Serotonergic and NA effects-same
- **α-1 blockade:** orthostatic hypotension, sexual dysfunction,
- **H-1 Blockade:** weight gain, sedation
- **Anti cholinergic:** dry mouth, Blurred vision, constipation, urinary hesitancy
- Sexual Dysfunction
- Block cardiac K channels: QT prolongation (TdP)
- Lowers seizure threshold: seizures may occur
- Overdoses TCA toxicity: 3 C's: Coma, Convulsions, Cardiac arrythmias (Rx- airway support, cardiac monitoring , gastric lavage and I.V $NaHCO_3$)

Atypical antidepressants

Bupropion

MOA- inhibits the reuptake of DA and NA into the neuron

Non-competivite antagonists of nicotinic acetylcholine receptors and may help block the reinforcing effects if nicotine

Indications:

- Smoking cessation effect- reduces the craving for nicotine
- Reversing SSRI- induced sexual dysfunction

A/E: Lowers seizure threshold (contraindicated in epilepsy)

5-HT2 Antagonists

Nefazodone, Trazodone

- Block 5-HT_2 potently (producing antidepressant, antianxiety & antipsychotic effects)
- α-1 receptor antagonist (causes orthostatic hypotension)

Trazodone

- Blocks 5-HT2
- Moderate - H1- receptor blockade (causes sedation)

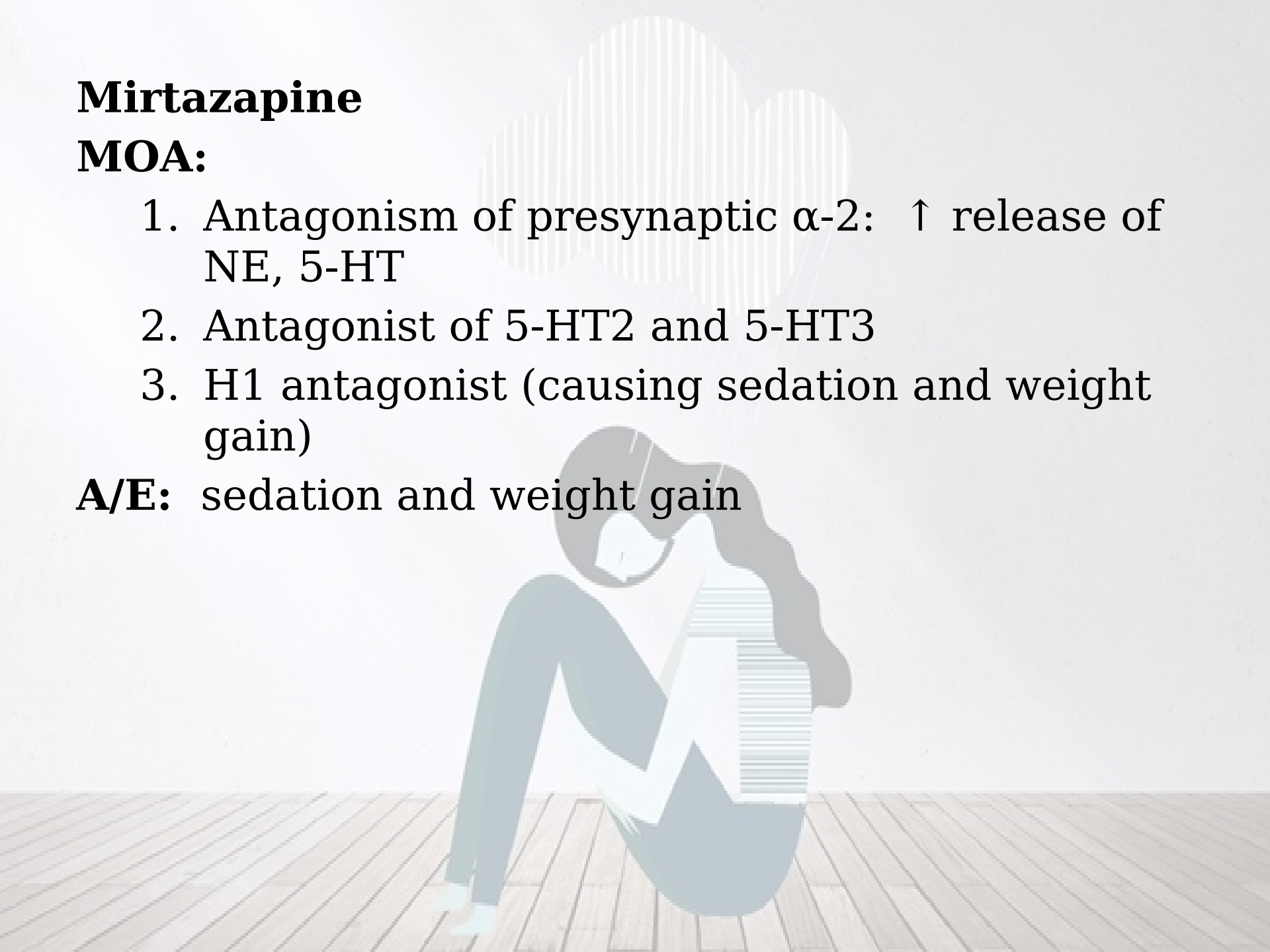
A/E: sedation and priapism

Mirtazapine

MOA:

1. Antagonism of presynaptic α-2: ↑ release of NE, 5-HT
2. Antagonist of 5-HT2 and 5-HT3
3. H1 antagonist (causing sedation and weight gain)

A/E: sedation and weight gain



Monoamine oxidase inhibitors (MAOIs)

MAO- mitochondrial enzyme involved in the metabolism of biogenic amines(Adr, NA, DA, 5HT)

2 isoforms-

MAO-A- metabolism of NA, 5HT & inhibited by **M**oclobemide

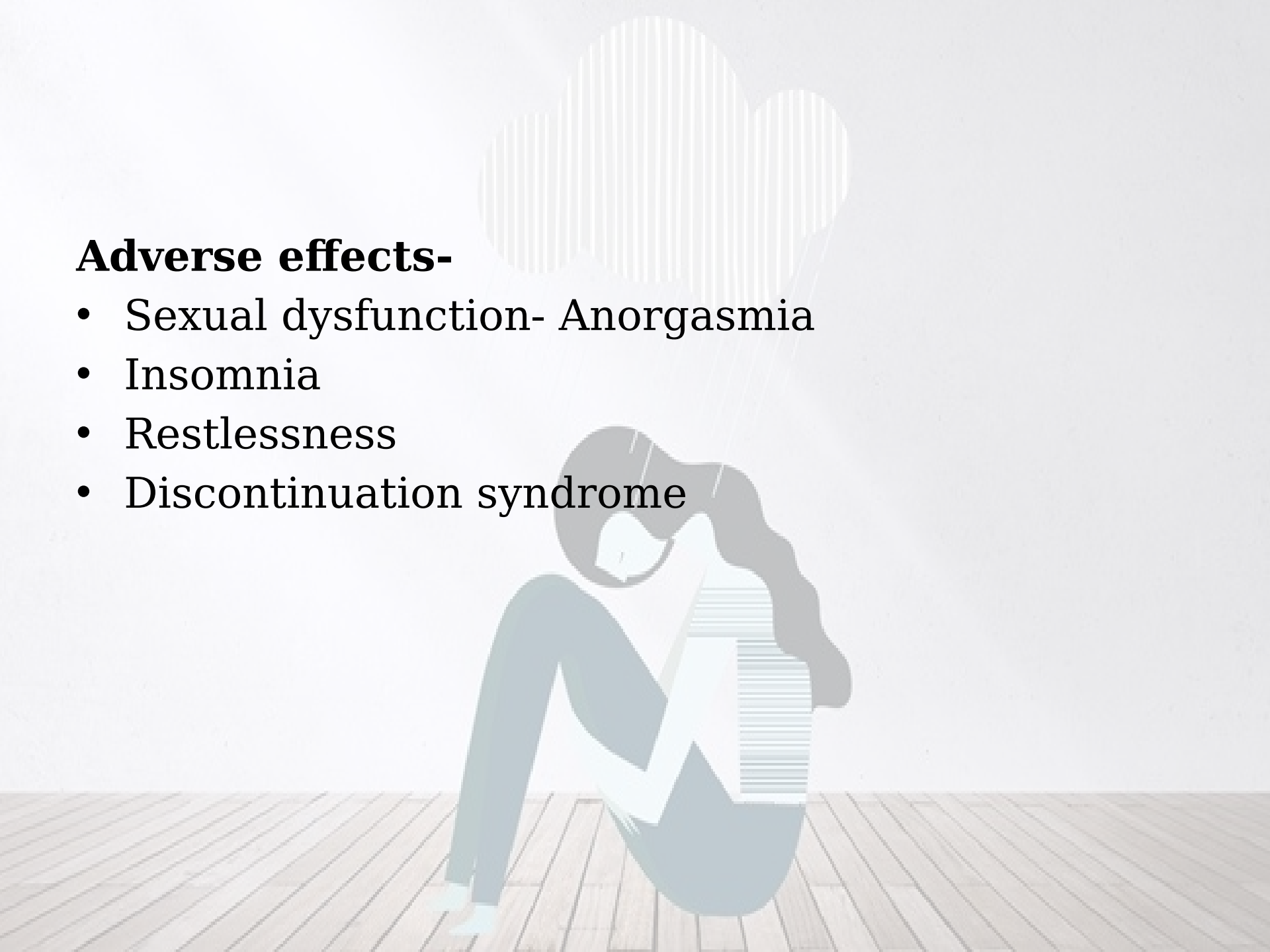
MAO-B- metabolism of dopamine & inhibited by **S**elegiline

Mechanism of action-

Increases the brain amine levels by inhibiting their metabolism in the nerve ending resulting in an increase in NE and 5HT

Adverse effects-

- Sexual dysfunction- Anorgasmia
- Insomnia
- Restlessness
- Discontinuation syndrome



Food interactions-

Cheese reaction:

MAOIs prevents the breakdown of tyramine in the gut

Food stuff rich in tyramine(aged cheese, beer, meat,red wine, yeast)

↓

reaches the systemic circulation

↓

Displaces large number of noradrenaline from sympathetic nerve endings

→

Leads to increase in BP → fatal hypertensive crisis and cerebrovascular accidents

Therapeutic Uses of Antidepressants

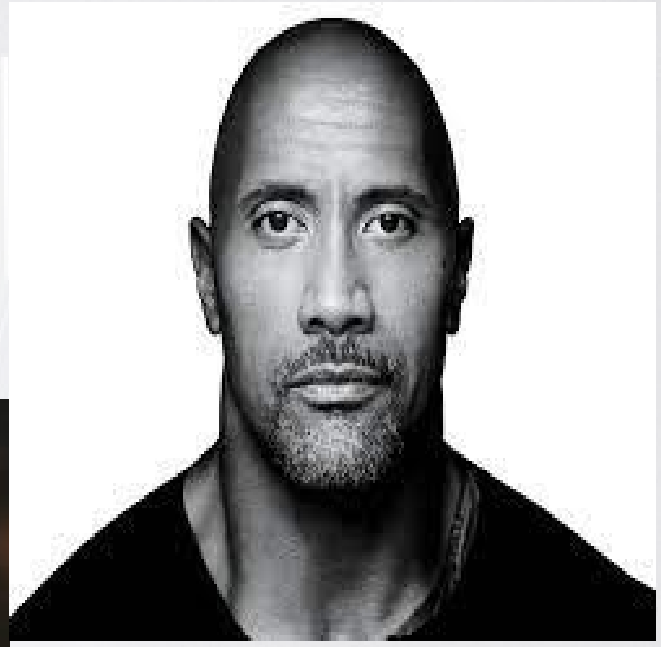
- **Major depressive disorder**
  - SSRIs, SNRIs, 5-HT antagonists, preferred over TCAs because of better tolerability
- **Bipolar disorder (for depressive phase)**
  - SSRI /TCAs in combination with Lithium (for maniac phase)
- **Panic disorder**
  - SSRIs, venlafaxine
- **Obsessive compulsive disorder-**
  - Clomipramine, SSRI (especially fluvoxamine)

Therapeutic Uses of Antidepressants

- **Enuresis (bedwetting)**
  - Imipramine
- **Attention deficit hyperactivity disorder(ADHD)**
  - TCAs and Atomoxetine
- **Nicotine dependence (to quit smoking)**
  - Bupropion
- **Neuropathic pain and fibromyalgia**
  - Duloxetine, TCA
- **GAD, panic attacks, social phobias, post-traumatic stress disorder, bulimia, and premenstrual dysphoric disorder**
  - SSRI

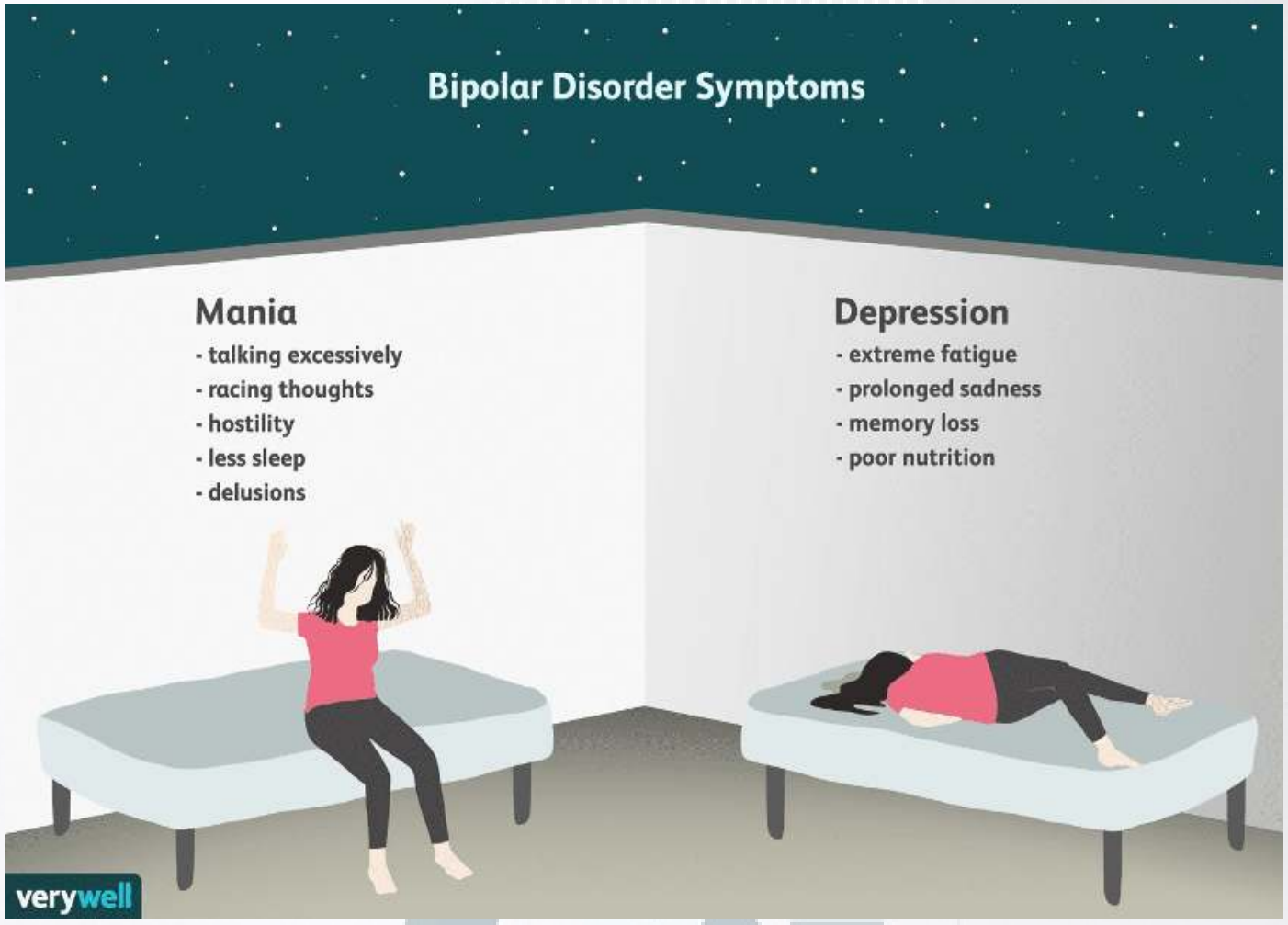











Bipolar Disorder Symptoms
Mania
- talking excessively
- racing thoughts
- hostility
- less sleep
- delusions
Depression
- extreme fatigue
- prolonged sadness
- memory loss
- poor nutrition
verywell

Drugs for the treatment of bipolar disorder

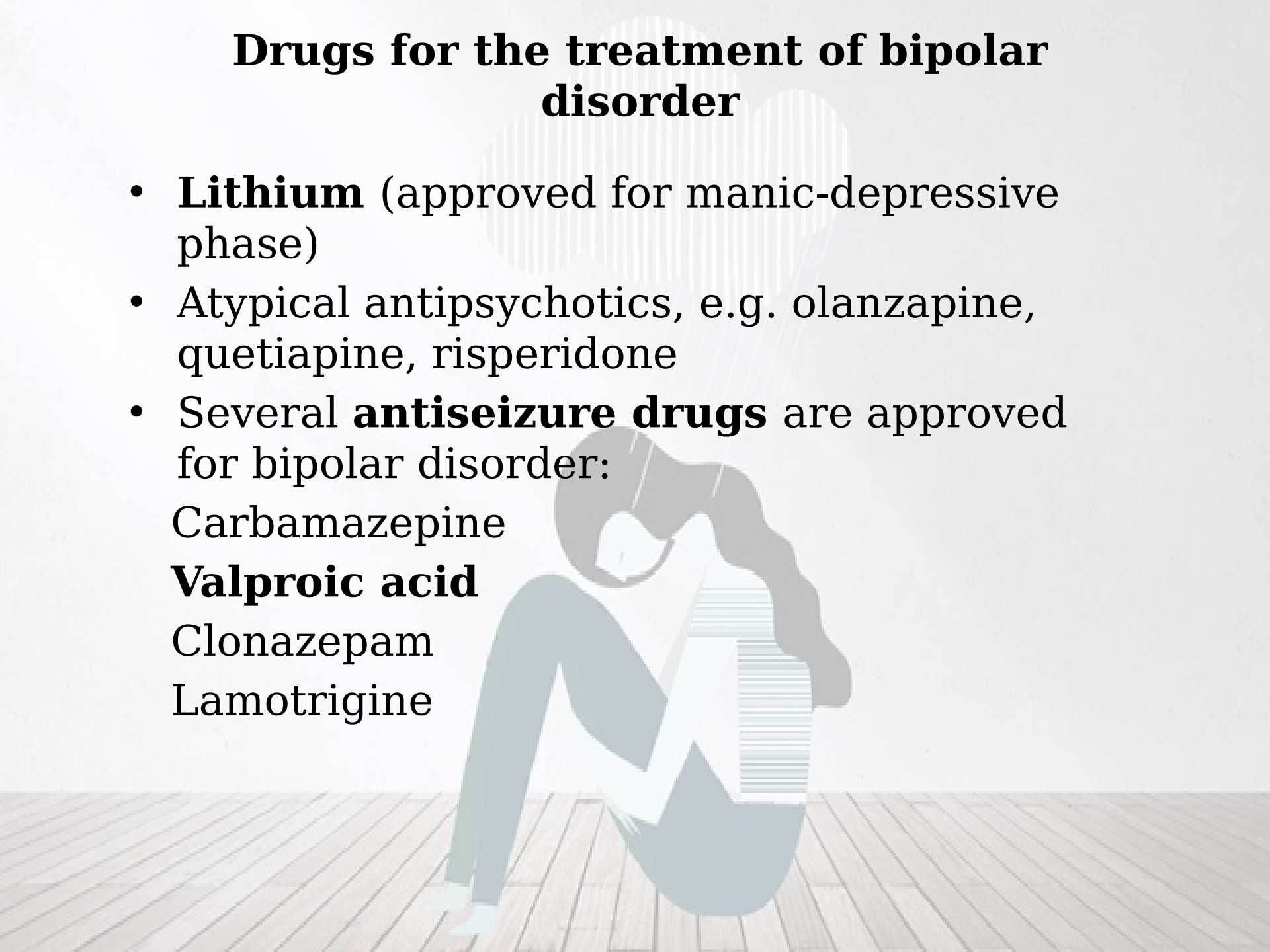
- **Lithium** (approved for manic-depressive phase)
- Atypical antipsychotics, e.g. olanzapine, quetiapine, risperidone
- Several **antiseizure drugs** are approved for bipolar disorder:

  Carbamazepine

  Valproic acid

  Clonazepam

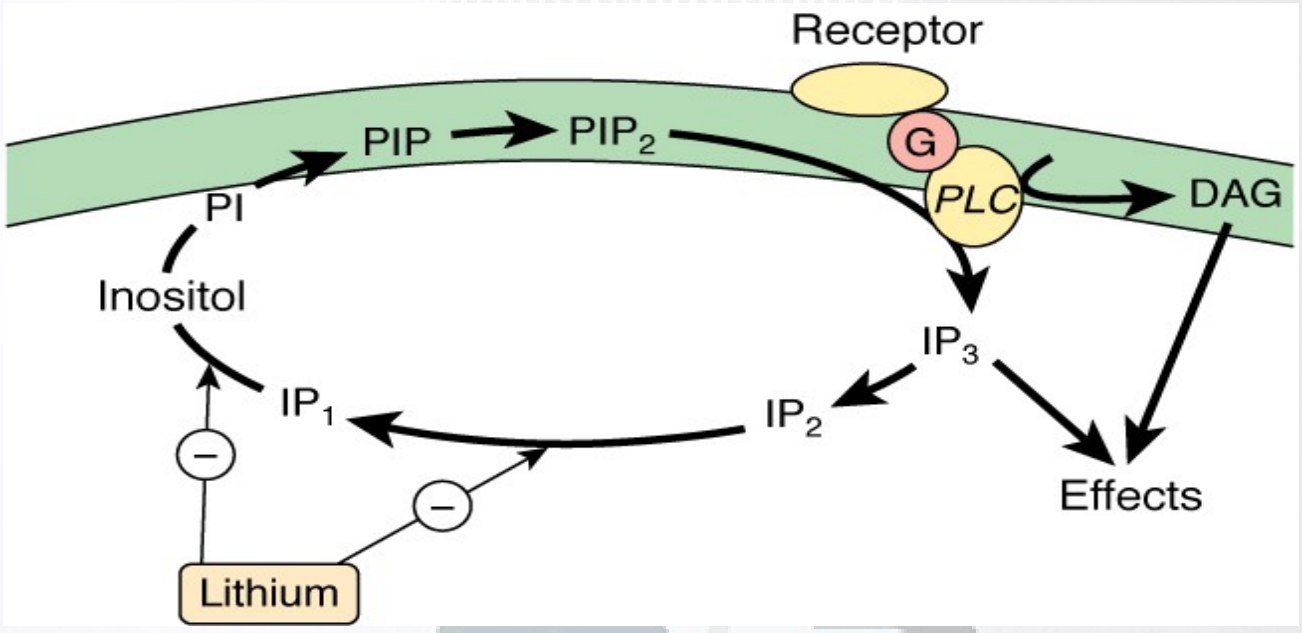
  Lamotrigine



Bipolar Affective Disorder-Lithium

- ✔ Considered the gold standard of mood stabilizers
- ✔ Is a small monovalent cation
- Long-term choice of treatment
  - Especially effective for mania
  - ~80% effective in acute mania
  - ~60% effective long-term
  - Being replaced by newer drugs
    - **Valproate (First choice in acute mania in US)**
    - Atypical antipsychotics

Effect of Lithium on the IP3 and DAG system



MOA: Lithium inhibits enzymes involved in the recycling of neuronal membrane phospholipids (IP, PIP & IP3)

IP3 – inositol triphosphate
DAG- diacyl glycerol

Pharmacokinetics

- $T_{1/2}$ - 20 hours
- The half life increases to 36hrs in elderly
- Excreted **almost entirely in the urine**
- Lithium has narrow therapeutic ratio
- Target plasma concentration is 0.8-1.2 mEq/L
- Levels should be checked 5 days after dose
- Plasma levels should be drawn 10-12 hours after the last dose

Lithium - Pharmacokinetics

Renal clearance



Lithium → **Excreted in urine**

Drugs that increase lithium toxicity
**Diuretics, NSAIDs, metronidazole, and ACE inhibitors
→ **Reduce clearance** → **Lithium toxicity**

Drugs that decrease lithium levels
**Acetazolamide, urea, xanthine preparations (including caffeine), and alkalinizing agents
→ **Increase clearance** → **Low dose of Lithium**

Adverse Effects of Lithium

- **Tremor (common), sedation, ataxia, aphasia**
  - ↓ w/ propranolol, atenolol
- **Reversible Nephrogenic Diabetes Insipidus**
  - Loss of response to ADH
  - Resistant to vasopressin
  - Treat w/ amiloride
- **Edema (common)**
  - Related to Na^+ retention
- **Depresses heart S-A node**
  - Avoid in “sick sinus” syndrome
- Use in pregnancy has been associated with **Ebstein’s anomaly**
- **Postpartum toxicity**
  - ↓Renal clearance after delivery
  - Transferred via breast milk

Valproic Acid

- First choice for the **treatment of acute mania** in US
- Used as a mood stabilizer
- Effective in lithium non-responders
- Effective in treating partial and primary generalized epilepsies
- Multiple mechanisms
  - Effect on ***Na^+ channels*** like that of phenytoin
  - Effect on ***T-type Ca^{2+} channels*** like that of ethosuximide
  - May increase concentration of GABA by ***stimulating GA-DC*** (Glutamic acid decorboxylase) enzyme
  - May also ***enhance K^+ channel*** permeability
- **Inhibits CYP enzymes** thus elevating the concentration of other drugs when given concomitantly (carbamazepine, lamotrigine, phenobarbital and phenytoin)
- Side effects may include somnolence, nausea, dizziness, and hepatic damage and teratogenicity
- Monitor liver function tests, platelets, and ammonia levels.

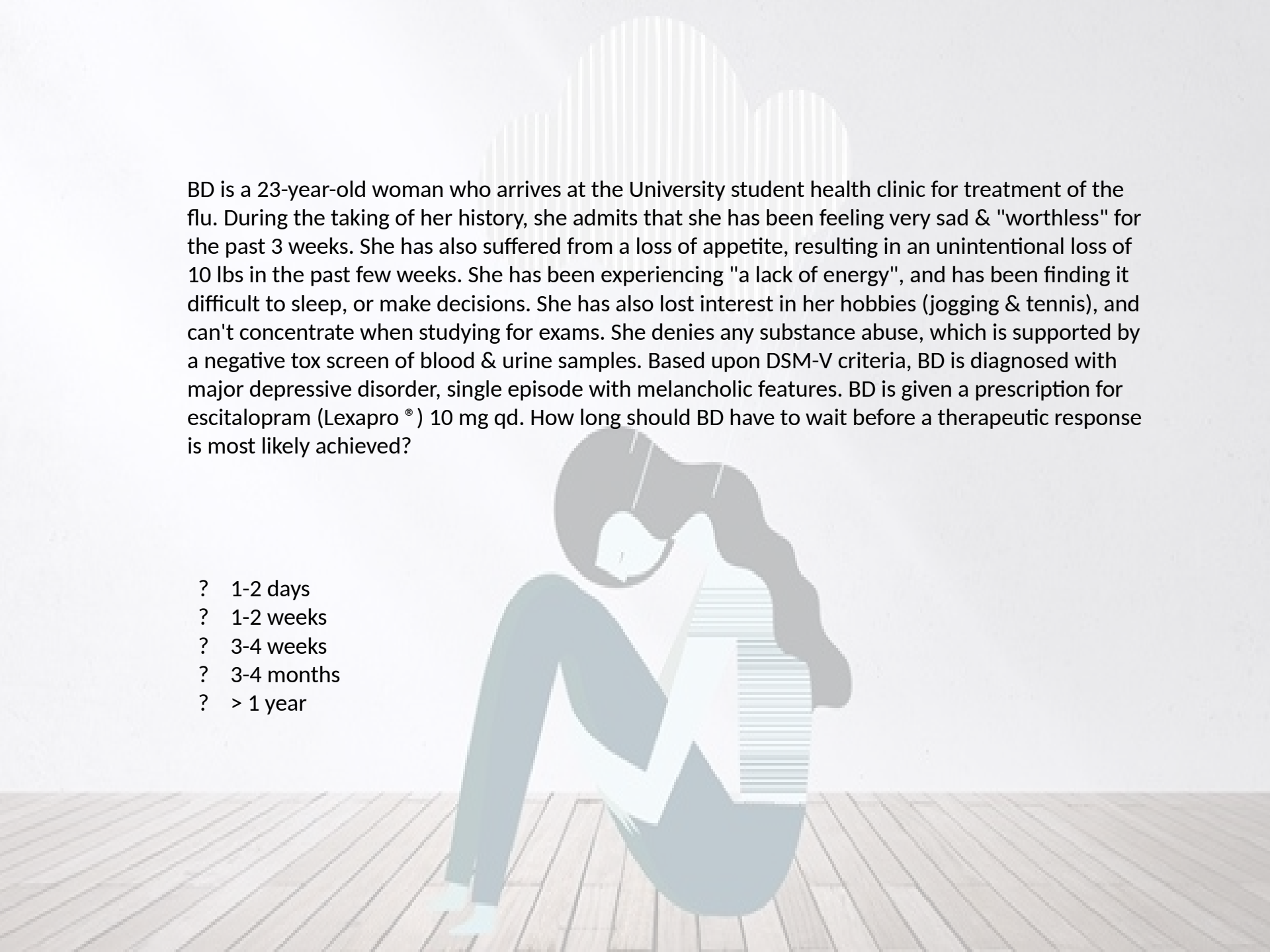
Lamotrigine

- Approved for maintenance treatment of bipolar disorder
- Appears particularly **effective in bipolar depression**
- Mechanism – inhibition of voltage-gated Na^+ channels, Ca^{2+} channels
- Must be titrated slowly due to the **risk of Stevens-Johnson syndrome**
- Co-administration with valproate may increase lamotrigine levels, and titration
- must be performed more slowly

Carbamazepine

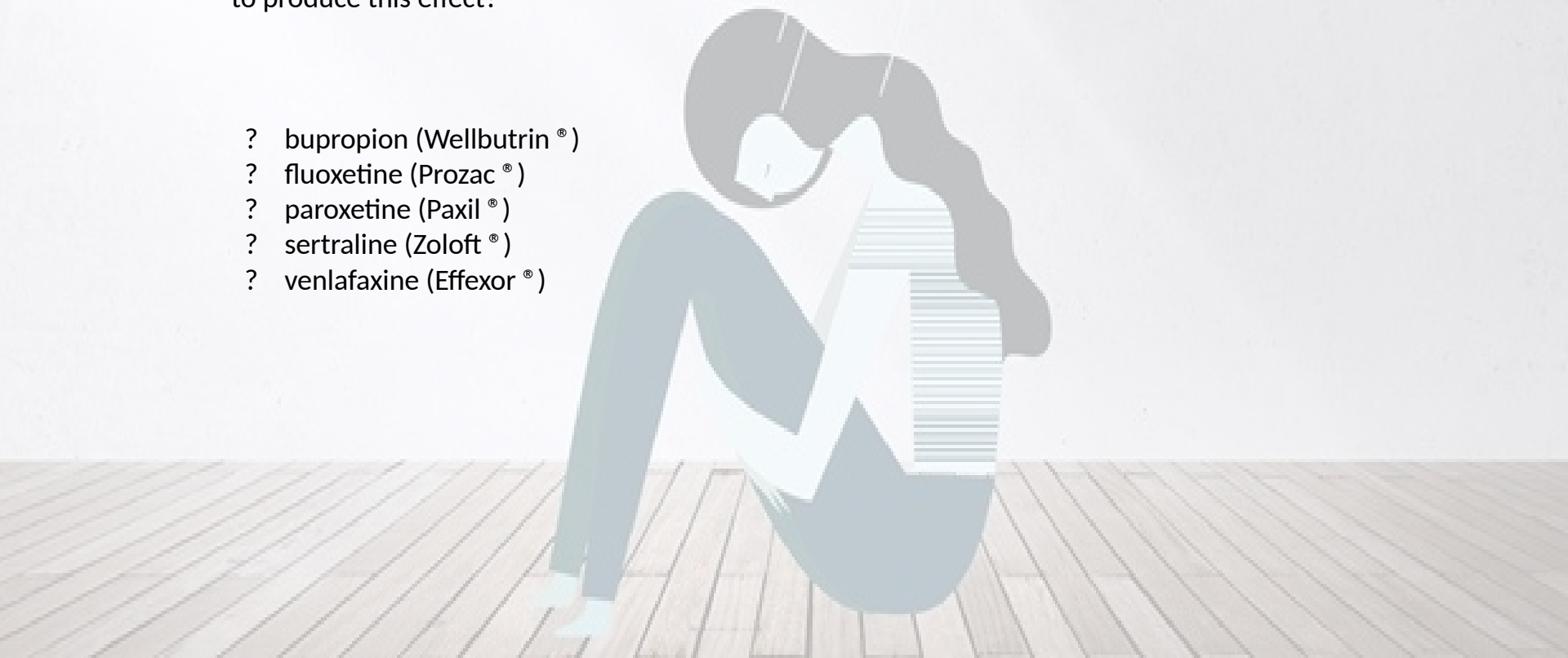
- **Primary drug for the treatment of partial and tonic-clonic seizures**
- May be used for acute mania and maintenance therapy, although these are not FDA-approved indications
- Block sodium channels inhibiting generation of repetitive action potentials
- Potential side effects include **agranulocytosis** and leukopenia
- Late complication of therapy: **Hyponatremia in elderly** patients especially with cardiac condition

BD is a 23-year-old woman who arrives at the University student health clinic for treatment of the flu. During the taking of her history, she admits that she has been feeling very sad & "worthless" for the past 3 weeks. She has also suffered from a loss of appetite, resulting in an unintentional loss of 10 lbs in the past few weeks. She has been experiencing "a lack of energy", and has been finding it difficult to sleep, or make decisions. She has also lost interest in her hobbies (jogging & tennis), and can't concentrate when studying for exams. She denies any substance abuse, which is supported by a negative tox screen of blood & urine samples. Based upon DSM-V criteria, BD is diagnosed with major depressive disorder, single episode with melancholic features. BD is given a prescription for escitalopram (Lexapro ®) 10 mg qd. How long should BD have to wait before a therapeutic response is most likely achieved?



? 1-2 days
? 1-2 weeks
? 3-4 weeks
? 3-4 months
? > 1 year

After 6 weeks, BD has a follow-up visit where she is asked about signs of improvement, as well as any side effects she has experienced. BD states that she has been sleeping better over the past 2 weeks. She recalls having mild transient bouts of nausea during the first week of treatment, but began taking her SSRI after breakfast, as instructed, and her nausea disappeared after several days. However she admits to having lost interest in having sexual relations with the fiancé, and recently has not been able to achieve orgasm (anorgasmia). You discuss the results of a clinical trial documenting improvement of sexual desire & reduced problems with achieving orgasm when a second drug is added. She elects to try augmentation with this drug. Which drug will you prescribe to produce this effect?



? bupropion (Wellbutrin ®)
? fluoxetine (Prozac ®)
? paroxetine (Paxil ®)
? sertraline (Zoloft ®)
? venlafaxine (Effexor ®)

Daniel is a 52-year-old man suffering from his first episode of major depression. He has a history of angina and survived a MI three months ago. Which agent would be contraindicated for treating his depression?

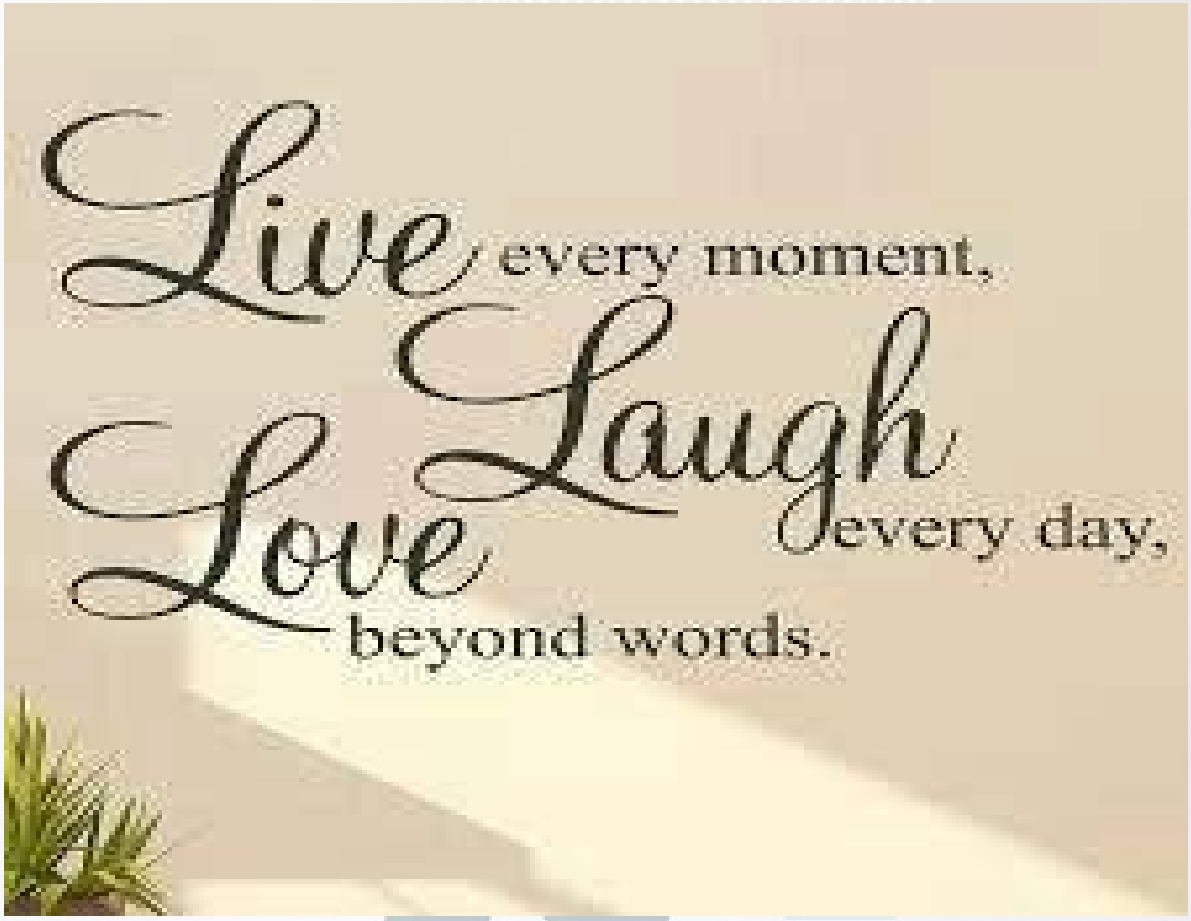
? amitriptyline
? bupropion
? duloxetine
? fluoxetine
? paroxetine



Jimmy T is a 56-year-old professor with a history of hypertension who recently underwent a coronary artery bypass graft (CABD) procedure. Jimmy has a past history of cigarette smoking (1/2 pack per day for the past 20 years). During the recovery phase of his open heart surgery, Jimmy is diagnosed with major depression and seeks treatment. Which antidepressant may be able to treat both his mental disorder & help him decrease his craving for cigarettes?

- ? bupropion
- ? desipramine
- ? fluoxetine
- ? nefaxodone
- ? sertraline





THANK YOU